

House Price Prediction Using Linear Regression

1. Introduction

Machine Learning is widely used for predictive analysis in various domains. In this project, a Linear Regression model was developed to predict house prices using the California Housing Dataset provided by Scikit-Learn.

The objective of this project is to understand the machine learning workflow, including data exploration, preprocessing, model training, evaluation, and visualization.

2. Dataset Description

The California Housing Dataset contains information about housing districts in California. The dataset includes the following features:

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

The target variable is the house price.

3. Exploratory Data Analysis (EDA)

The dataset was analyzed using Pandas functions such as:

- `df.head()`
- `df.info()`
- `df.describe()`
- `df.isnull().sum()`

Observations:

- The dataset contains 20,640 records.
- There are 8 input features and 1 target variable.
- No missing values were found in the dataset.
- Statistical summaries were generated for all features.

A correlation heatmap was also created to understand the relationships between variables.

4. Data Preprocessing

The dataset was divided into features (X) and target variable (y).

The data was split into:

- Training Set: 80%
- Testing Set: 20%

Training Shape: (16512, 8)

Testing Shape: (4128, 8)

The `train_test_split()` function from Scikit-Learn was used with `random_state = 42`.

5. Model Building

A Linear Regression model was implemented using Scikit-Learn.

Steps performed:

1. Created a Linear Regression object.
2. Trained the model using training data.
3. Generated predictions using test data.
4. Evaluated model performance.

6. Model Evaluation

The performance of the model was evaluated using three metrics:

Mean Absolute Error (MAE)

MAE = 0.5332

Root Mean Squared Error (RMSE)

RMSE = 0.7456

R² Score

R² Score = 0.5758

These results indicate that the model has moderate predictive capability and can reasonably estimate house prices from the provided features.

7. Visualizations

The following visualizations were generated:

1. Correlation Heatmap
2. Actual vs Predicted House Price Scatter Plot

These visualizations helped in understanding feature relationships and model performance.

8. Conclusion

A House Price Prediction model was successfully developed using Linear Regression. The California Housing Dataset was explored and analyzed, and the trained model achieved satisfactory performance with an R^2 Score of 0.5758.

The project provided practical experience in data preprocessing, exploratory data analysis, machine learning model development, evaluation, and visualization.

9. Future Improvements

Future enhancements may include:

- Feature Engineering
- Hyperparameter Tuning
- Random Forest Regression
- Gradient Boosting Models
- XGBoost
- Deep Learning Approaches

10. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Jupyter Notebook

Author

Viraj Panthi

B.Tech CS (AI & ML)